

Fracture Aperture Measurements and Migration of Solutes, Viruses, and Immiscible Creosote in a Column of Clay-Rich Till

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Abstract

A series of ground-water flow and tracer experiments were performed on an undisturbed column of fractured clay-rich till, 0.5 m diameter by 0.5 m long, in a pressure-controlled cell. The measured hydraulic conductivity of the sample was 1.0 to 1.2×10^{-6} m/sec and the average hydraulic gradient during the tracer experiments ranged from 0.45 to 0.49. The experiments clearly show that ground-water flow and contaminant migration through the sample is primarily controlled by fractures and root holes. Tracer experiments using a solute (chloride), colloid-sized bacteriophage (PRD-1 and MS-2) and uncharged latex microspheres, indicated very fast transport rates of 4 to 360 m/day. These rates are similar to fracture flow velocities calculated on the basis of the measured bulk hydraulic conductivity of the column, and measured fracture spacing, using the cubic law for flow through parallel-walled fractures. Fracture aperture values calculated from the ground-water flow data (35 to 56 μm) are of the same magnitude as values calculated from the breakthrough of tracers (13 to 120 μm). Aperture values calculated for fractures (1 to 94 μm) and root holes (2 to 188 μm), on the basis of measured immiscible creosote entry pressures, are also comparable with these values. The injected creosote, a DNAPL, penetrated most of the visible and a few invisible fractures and root holes, indicating that, for this till, fractures and root holes are important conduits for the transport of DNAPL's.

Introduction

Fractures in clay-rich tills can greatly increase the hydraulic conductivity of such deposits and hence increase the potential for migration of contaminants (e.g., McKay et al., 1993a, b; Jørgensen and Fredericia, 1992; Fredericia, 1990). The fracture orientation, length, degree of interconnection and the aperture, or width of the opening, are critical factors in controlling hydraulic conductivity and influencing the migration of contaminants. Fracture orientation, length, and interconnection can often be observed and directly measured in excavations or in core samples, particularly if there is oxidation staining along the fractures. Fracture aperture values in a deposit are expected to vary significantly, both from fracture to fracture and within the same fracture (Tsang, 1984). They cannot be directly measured in most clays because of their small size, a few micrometers to a few hundreds of micrometers, and because they are often influenced by disturbance during sampling or excavation (McKay et al., 1993a). Aperture values are generally determined indirectly from measurements of hydraulic conductivity and spacing of visibly stained fractures, using the "cubic law" which is based on an analogy to flow between smooth parallel plates (Snow, 1969). The calculated aperture value represents an "average" aperture for the

volume of till influenced by the hydraulic conductivity measurement. Because of all these assumptions and the difficulty in obtaining reliable measurements of hydraulic conductivity in till there is always a large degree of uncertainty associated with values of "hydraulic-aperture" (McKay et al., 1993a).

Discrete-fracture ground-water flow and contaminant transport models (e.g., Sudicky and McLaren, 1992) generally employ the cubic law for determining flow and require input data on the geometry of the fracture network and values of fracture aperture. Uncertainty in these values greatly influences evaluation of contaminant transport. Grisak et al. (1980a and b) simulated chloride transport in a column of fractured clay till using a discrete-fracture model based on the cubic law, which also included diffusion from the fractures into the porous matrix. To fit the simulations to the data they varied both fracture spacing and aperture, but held the total flow through the sample equal to the measured constant flow value. The best-fit was achieved with a fracture spacing of 12 cm, which was a factor of three larger than the measured fracture spacing of 4 cm. In a recent field experiment in fractured clay tills in Canada, McKay et al. (1993b) compared values of "hydraulic-aperture" with aperture values derived from a discrete-fracture/matrix-diffusion type model (similar to the type used by Grisak et al., 1980a and b), which was fitted to measured bromide breakthrough curves. They found that values derived from the two methods, using the same fracture spacing, were generally within a factor of two, which, although of the same magnitude, correspond to substantial differences in flow rates. They also injected two strains of colloid-sized bacteriophage tracers which, because of their large diameter relative to the pores in the blocks between fractures, were not expected to be retarded by diffusion into these pores and hence were used to measure flow velocity. The flow velocity in the largest aperture fractures, as indicated by first arrival of the bacteriophage, was 2 to >5 m/day, which fell in between most

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of the velocity values calculated from the hydraulic conductivity measurements and the solute tracer data.

The above studies indicate that using calculated values of “hydraulic-aperture” and flow velocity for predicting solute transport may be acceptable in some, but not all, settings. As well, because the cubic law was involved in all of the calculations of fracture aperture, there was no independent method of assessing the validity of the values. This is important because aperture size is a critical factor in controlling movement of other types of commonly occurring contaminants, particularly Dense Non-Aqueous Phase Liquids, DNAPL's, (Kueper and McWhorter, 1991). Since entry of DNAPL's will tend to be controlled by the larger aperture fractures it is important to determine if calculated “hydraulic-aperture” values are appropriate for assessing DNAPL behavior.

The main objective of this paper is to present fracture aperture values for a sample of fractured Danish till calculated using (1) hydraulic data, (2) solute-transport data, (3) colloid-transport data, and (4) measurements of nonwetting fluid entry pressure for a dense, immiscible creosote. The distribution of creosote in the fractures is also observed to determine if certain fractures are more readily penetrated by the creosote and hence have larger aperture values. A secondary objective is to determine whether colloid-sized tracers are sufficiently mobile in the Danish tills to be used as indicators of fracture flow velocity in field-scale experiments.

Field Site and Geology

The till column sample for the tracer experiments was collected at an orchard, 0.5 km west of Skælskør in western Sjælland, Denmark (Figure 1). The orchard has an area of approximately 0.5 km², and is situated in a moderately undulating till plain which overlies Danien-Maastrichtien (late Cretaceous-early Paleocene) bryozoan limestone. Core samples from the site suggest that the 20 to 25 m thick till sequence consists of different stratigraphic units, the uppermost of which was deposited during

the last glaciation (Weichsel) of Northern Europe approximately 13–75 thousand years ago (Houmark-Nielsen, 1994). There is good natural drainage at the site due to topographic relief and infiltration. The water table fluctuates annually between 1.3 and 4.6 m below ground surface. The lithology and structure of the upper 6 m of the weichselian till at the site were mapped and studied in detail, and are summarized as follows. The upper 4.6 m of the till is oxidized and overlies a 0.3 to 0.5 m thick primarily unoxidized transition zone, which leads into the unoxidized till at greater depths. In the oxidized zone, and in the transition zone, hydraulically active fractures are easily recognized due to red, black, and grey oxidation/reduction staining along the fractures. Near-vertical and horizontal fractures were recognized throughout the profile during collection of the column sample. The spacing between the vertical fractures (distance between fractures measured on a horizontal scan line) increases from approximately 5 cm, in the upper 2 m of the profile, to 20 cm at 6 m below the surface. Horizontal fractures, with a spacing of up to 5 cm, were dominant in the unoxidized zone, but occurred less frequently in the oxidized zone. Root holes and worm burrows were observed to depths of 2.5 and 1.8 m, respectively, and both were found only in the oxidized zone.

Material and Methods

Intact Soil Column and Experimental Setup

A cylindrical sample of undisturbed till, 0.5 m diameter by 0.5 m in length, was taken from 2–2.5 m depth in a test pit at the field site (Figure 1). The soil at this depth is a massive clay-rich till which is highly weathered and oxidized to a light brown color. The sample contained many near-vertical fractures, which were identified mainly by the presence of a “halo” of grey-brown reduction staining around the fractures. Typical fracture spacings were approximately 5 to 10 cm and typical fracture lengths, measured on the sample or on the exposed excavation surface, were 5 to 30 cm. A few fractures appeared to penetrate the entire sample. There were also many fine (< 1 mm diameter) root holes or imprints along the fractures and a few in the matrix blocks between fractures. A grain-size distribution for a subsample of the till column was determined using standard sieving and hydrometer methods. The sample contained 12% clay, 31% silt, and 57% sand and gravel.

The methods used for collecting the sample and setting up the experiment are based on those described by Jørgensen and Spliid (1992) and Jørgensen and Foged (1994) and are briefly described as follows. The sample was excavated with hand tools and an expandable steel casing was advanced over the sample as it was trimmed to its final diameter. A flexible rubber membrane was placed between the sample and the casing with fluid rubber injected between the sample and the membrane. Before hardening the fluid rubber entered a few millimeters of the till, providing a seal between the soil and the membrane. The base of the column was then cut free from the underlying till and temporary steel end caps were attached for transport.

In the laboratory, the steel casing was removed and permanent end caps, with sampling ports, were put in place and bonded to the rubber liner. The sample was then placed in a large steel pressure cell (Figure 2) in an inverted position. The pressure cell around the sample was filled with water and a confining pressure of 45 kPa was applied to the sample using a tank of compressed gas and a pressure regulator. The confining pressure was maintained during the solute and colloid tracer

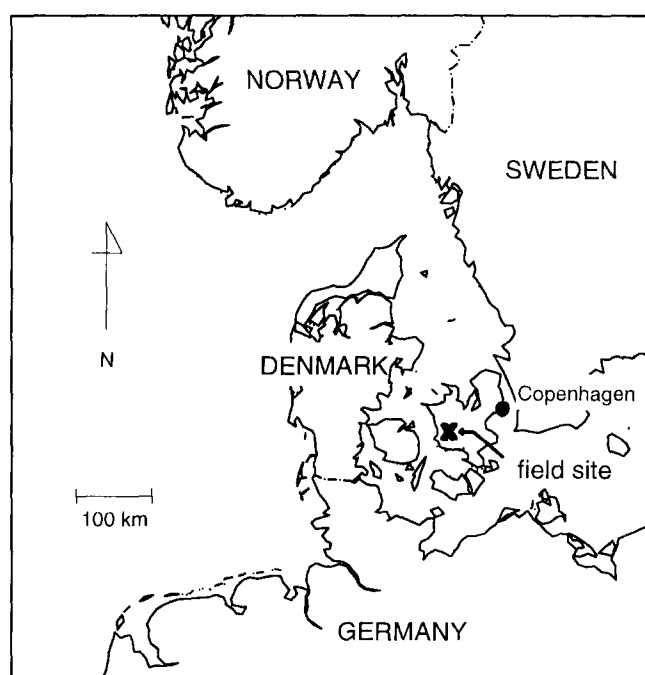


Fig. 1. Denmark, with location of field site indicated.

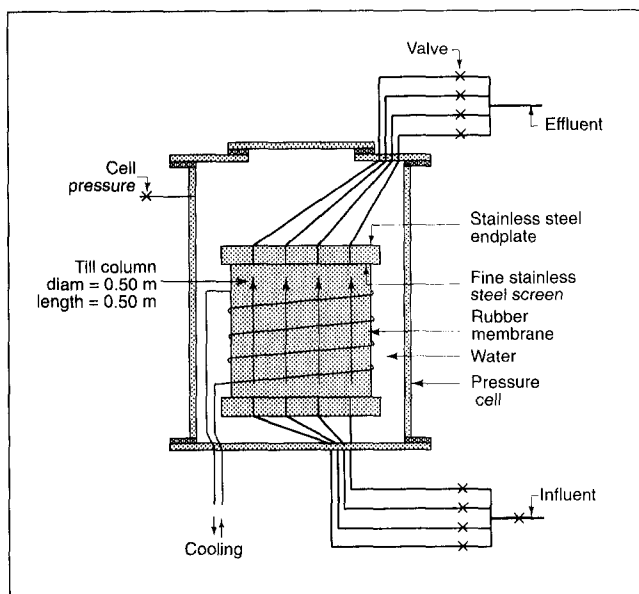


Fig. 2. Schematic illustration of the experimental setup for the solute and colloid tracer injections.

experiments. The confining pressure corresponds to the expected in situ pressure for the sample. The sample was then saturated by flow from the bottom end cap. Since the sample was inverted, this was analogous to vertical downward flow in a natural setting and had the advantage of allowing air bubbles to escape through the top. The sample was then cooled to about 8°C, the approximate in situ soil temperature, by means of a closed-loop system which circulated antifreeze through the pressure cell (Figure 2).

A 40 liter reservoir, which was also cooled to about 8°C, was connected to the system. The hydraulic gradient across the sample was maintained at approximately 0.5 by periodically adjusting the elevation of the reservoir and/or the effluent line to maintain a constant elevation difference between the surface of the reservoir and the discharge point of the effluent line. The flow through the sample was monitored by periodic direct measurements of the effluent discharge rate using a graduated cylinder and a stopwatch.

The water used for influent was obtained from a well in a confined aquifer at a site that is geologically similar to the site where the column was taken. The chemical composition (major ions) of the influent water before addition of the tracers, was in mg/l: TDS 730; Ca²⁺ 120; Mg²⁺ 31; Na⁺ 31; HCO₃⁻ 408; SO₄²⁻ 73; Cl⁻ 72; and the pH was 7.4. After addition of the bacteriophage solution to the ground water, the chloride concentration of the mixture increased about 20 mg/l with an approximate corresponding increase in the sodium concentration.

Tracer Experiments

Three separate tracer injections were carried out over a period of approximately two weeks. The first injection involved adding two strains of colloid-sized bacteriophage (viruses that replicate in bacteria), PRD-1 and MS-2, with a diameter of 0.062 and 0.026 µm respectively, and 1 ml of a solution containing 0.10 µm diameter carboxylated latex microspheres to the ground water in the influent reservoir. The reservoir was then stirred and the resulting tracer concentrations were 6.0×10^7 pfu/ml (plaque forming units per milliliter) of PRD-1, 2.9×10^5 pfu/ml of MS-2 and 1.92×10^3 microspheres/ml. Before the tracers

were added all of the valves on the influent and effluent lines were closed. Approximately 2 liters of tracer solution were then flushed through the influent lines and the end cap of the sample over a 15 minute period. This procedure was used to try to minimize the influence of storage volume of the lines and end cap. The valves on the effluent line were then opened to start the tracer experiment. The first injection continued for approximately 48 hours. The tracer reservoir was then replaced with another reservoir containing clean ground water (no tracers added) and flushing of the colloidal tracers from the sample was monitored.

Samples of the effluent, and of the influent reservoir, were taken in 10 ml or 50 ml sterile plastic vials throughout the tracer experiments. The vials were kept refrigerated until all of the tracer analyses were complete.

The second injection began after 260 hours of flushing the sample with ground water containing no colloidal tracers. The valves on the influent and effluent lines were closed and the tracers (PRD-1 and MS-2) were added to the reservoir. The resulting tracer concentrations in the reservoir were 2.0×10^8 pfu/ml of PRD-1 and 1.8×10^3 pfu/ml of MS-2. Before the injection began, tracer solution was flushed through the main influent line, but not through the end cap. All of the valves were then opened and the second injection began.

The third injection, using chloride as a tracer, began approximately 7 hours later, while the second bacteriophage injection was still in progress. The influent and effluent valves were briefly closed and 100 ml of a concentrated solution of CaCl₂ was added to the reservoir which was then stirred. The resulting concentration of chloride in the reservoir was 490 mg/l compared to a background concentration of 92 mg/l immediately before the calcium chloride was added. The influent line was then flushed and the valves opened in the same manner as described for the second bacteriophage injection. After approximately 43 hours of injection of chloride, the reservoir was replaced with another reservoir containing clean ground water.

Description of Tracers and Analytical Methods

The bacteriophage PRD-1 and MS-2 were used as colloidal tracers in both experiments and polymer microspheres were also used in the first experiment. PRD-1 is a 0.062 µm diameter, icosahedral plasmid-dependent DNA bacteriophage (Olsen et al., 1974) obtained from Dr. Junetsu Ito (Department of Microbiology and Immunology, College of Medicine, University of Arizona, Tucson, AZ). The host bacteria for PRD-1 is *Salmonella typhimurium* LT-2. MS-2 coliphage is an icosahedral single-stranded RNA bacteriophage with an average diameter of 0.026 µm, which was obtained from the University of Arizona, Department of Microbiology and Immunology culture collection. The host bacteria for MS-2 is *Escherichia coli* (ATCC 15597).

Bacteriophage assays were performed at the Geological Survey of Denmark in Copenhagen with assistance from staff of the Department of Microbiology and Immunology at the University of Arizona (Tucson). The bacteriophages were assayed by the plaque-forming-unit method as described by Adams (1959) and Bales et al. (1991). Serial dilutions using Tris buffered saline and 1 ml of water from each sample were prepared and added to test tubes that contained 3 ml of molten overlay agar and 1 ml of 3-6 hour culture of *S. typhimurium* LT-2 and *E. coli* (ATCC 15597), and the mixture was then poured onto solidified Typtic Soy Agar plates. The plates were incubated overnight at

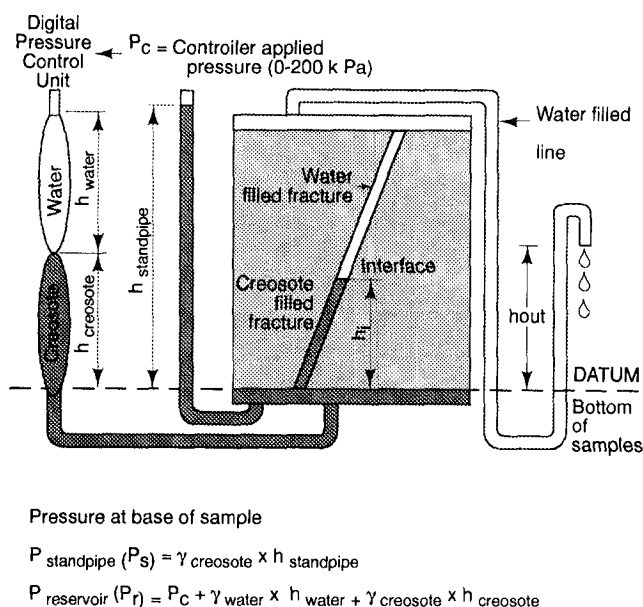


Fig. 3. Schematic illustration of the experimental setup for the immiscible creosote injection.

37°C, then counted using standard techniques. Most of the assays were completed within 24 hours of sample collection.

The microspheres used in the first bacteriophage injection were 0.10 μm diameter Fluoresbrite Carboxylate Microspheres (Polysciences, Inc., Warrington, PA, U.S.A.). Microsphere concentrations in effluent water were determined by filtering samples through polycarbonate membrane filters and then counting the beads using an epifluorescent microscope.

The dissolved chloride was analyzed by flow injection analysis using mercury (II) thiocyanate. An in-line electrical conductivity electrode attached to the effluent line of the column was used to assist in determining which samples were to be analyzed for chloride content.

Experimental Setup for Creosote Injection

An immiscible creosote mixture was injected into the till column over a one-week period. The experimental setup is shown on Figure 3. The creosote used was a mixture of two different commercial products obtained from Tarconord, Nyborg, Denmark: a very viscous black coke-oven coal tar (95% of mixture) and a solvent (5% of mixture), that is the light fraction of the distillate from coal tar. The density and the interfacial tension between the creosote and water were measured at 8°C for the two creosote products and for the resulting mixture. The viscosity of the solvent was also measured at 8°C. For the coal tar and the creosote mixture the viscosity was measured at temperatures between 18 and 38°C and then the viscosity at 8°C was estimated

from the determined exponential viscosity-temperature function. The density, interfacial tension, and the viscosity of the fluids were measured at the Technical University of Denmark (Chemical Engineering Department) by pycnometry, "drop volume" and viscometry respectively, with a standard deviation of less than 0.5%, 1%, and 3.3% on the results from these methods. The physical properties of the fluids are shown in Table 1.

The creosote mixture was placed in a stainless steel reservoir attached to a digital pressure/flow controller (Figure 3), and the creosote reservoir was connected to the influent lines at the base of the sample. The controller was preset to apply a constant pressure (measured in kPa) to the creosote mixture, and the flow was then measured by the same device and recorded by a printer. A second pressure measuring device, consisting of a clear standpipe connected to the sample end cap, was installed to directly measure pressure at the base of the sample. The pressure in the reservoir was raised in small increments until the lines were filled and the flow of creosote was stopped by the entry pressure necessary to enter the sample. The pressure was then raised further in small increments until initial flow into the sample was observed. The flow was monitored during the entire creosote injection and when flow slowed down or stopped, the creosote pressure was raised again. This stepwise procedure was repeated several times over a one-week period. As the creosote pressure was increased for the final injection of creosote, the confining pressure in the cell was also increased by an amount corresponding to the expected increase of the standpipe pressure (P_s) as estimated from the relative increase of the reservoir pressure (P_r) (see Figure 3). This was done to prevent leakage between the sample and the membrane.

Results and Discussion

Hydraulic Conductivity

The hydraulic gradient and the flow rate during the tracer experiment varied slightly as shown on Figures 4a and 4b. During injection 1 the mean hydraulic gradient was 0.49 and the mean flow rate was 6.0 ml/min. During injections 2 and 3 (combined) the mean values were 0.45 and 6.6 ml/min. The bulk hydraulic conductivity values calculated using the mean gradients and mean flow rates for injection 1 and injections 2 and 3 (combined) were 1.0×10^{-6} and 1.2×10^{-6} m/sec, respectively.

The variations in flow rate and hydraulic conductivity during the experiment are believed to be partly due to redistribution of fine-grained soil particles around the influent and effluent ports. Gas bubbles at the effluent ports and possible clogging of fractures due to the colloidal tracers may also have contributed to the variability. When flow rates declined significantly the operator would temporarily close three of the effluent ports, forcing all of the flow through the remaining port. When this was done gas bubbles and a small amount of sediment were

Table 1. Physical Properties of the Two Mixed Creosote Products and the Injected Creosote Mixture

	Density (kg/l)	Specific weight (N/m ³)	Dynamic viscosity (cp) ^a	Interfacial tension (N/m)
Coal tar	1.16	11.4×10^3	4.66×10^3 ^b	0.0197
Solvent	0.96	9.41×10^3	47.2	0.0174
Creosote mixture	1.15	11.3×10^3	1.14×10^3 ^b	0.0181

^a1cp = 10^{-3} kg/m/s.

^bExtrapolated from measurements between 18 and 38°C; all other parameters measured at 8°C.

often observed in the discharge. It is not known whether the bubbles came from air introduced when the sample was excavated, or whether they were formed later by biological action.

There were no field-measured values of hydraulic conductivity from the site to compare these results with, but the laboratory-measured values are within the range (although near the upper end) of field-measured values in similar Danish tills (Fredericia, 1990). Based on this comparison it is believed that the sample is hydraulically representative of a typical Danish till and that the influence of disturbance during sampling is relatively small. The hydraulic gradient of 0.45–0.49 is also within the range of vertical gradients, typically observed in till deposits in Denmark.

Transport of Colloids

During the first colloid injection (Figures 5a and 5b), both the bacteriophage PRD-1 and MS-2 and the microspheres were detected in the effluent within 4 minutes of the start of the injection. There were no detectable bacteriophage or microspheres in any of the background samples prior to the injection. The concentration of PRD-1 in the effluent increased rapidly until about 9 minutes after the start of the injection. The rate of increase then slowed greatly and approached a near steady-state concentration within 1 hour. The steady-state concentration of PRD-1 was approximately 2×10^7 pfu/ml, which is a factor of

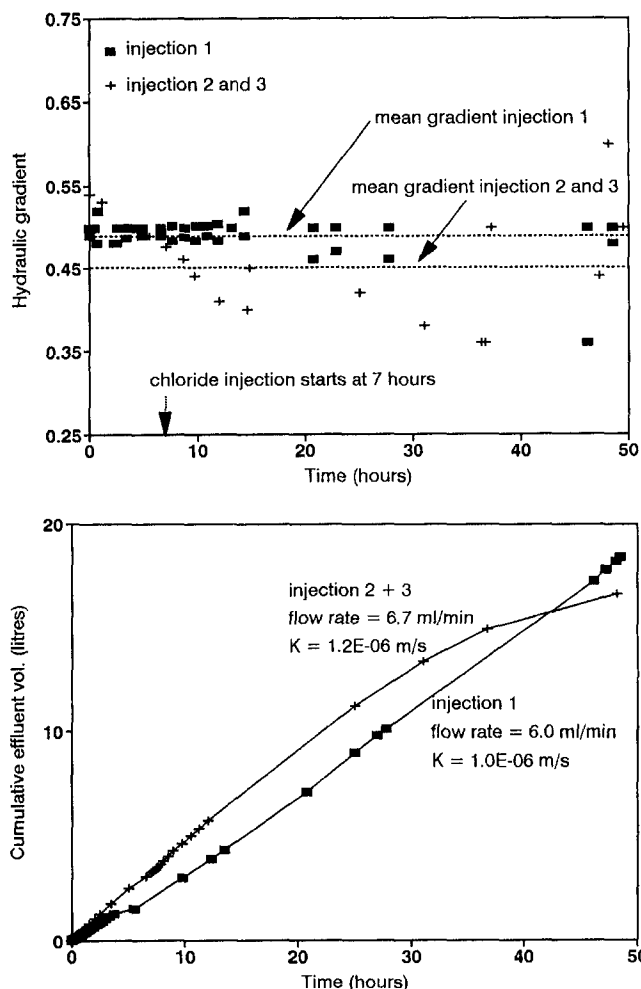


Fig. 4. Hydraulic conditions during the solute and colloid injections: (a) Hydraulic gradient as a function of time; (b) Cumulative effluent volume, average flow rate, and hydraulic conductivity.

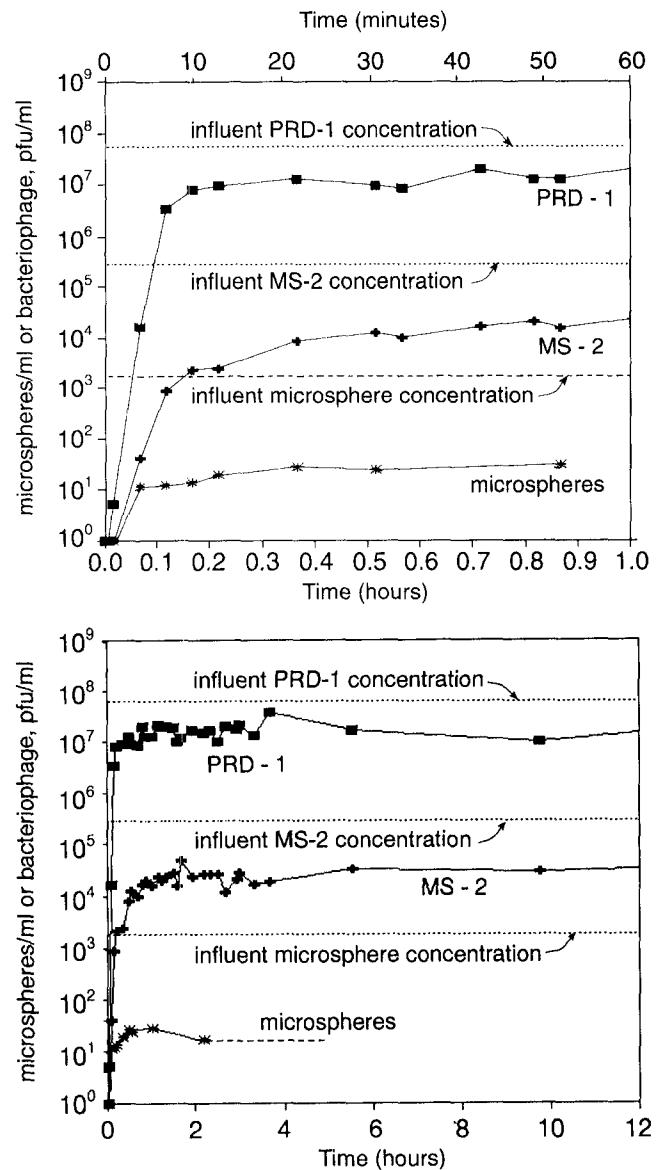


Fig. 5. Concentration of bacteriophage and microspheres in effluent, experiment 1. (a) 0 to 1.0 hours; (b) 0 to 12 hours.

three lower than the injection concentration. The concentration of MS-2 and the microspheres followed much the same trend. The steady-state concentration of MS-2 and the microspheres was approximately 3×10^4 pfu/ml and 25 microspheres/ml, which is one order and two orders of magnitude lower than the injection concentration, respectively.

The breakthrough of bacteriophage during the first injection was very fast and may have been influenced by the flushing of the end cap with tracer solution a few minutes before the test began. This could have caused some premature circulation of bacteriophage in the sample. For this reason, the bacteriophages were subsequently flushed out of the sample and the injection repeated. The results from the second injection are described below.

The breakthroughs of PRD-1 and MS-2 during the second injection follow much the same trend as observed in the first injection. This indicates that the flushing of the end cap before the first injection had relatively little influence, and confirms that colloid transport rates through the column are very fast. The times of first arrival of the tracers following the second injection were approximately 6 and 12 minutes for PRD-1 and MS-2,

respectively (Figures 6a and 6b). The time of first arrival was more difficult to measure for the second injection because for both tracers there were some bacteriophages remaining from the first injection. Concentrations of both bacteriophage tracers approached a near steady-state value within one to three hours of the start of the second injection. All of the above arrival times include the transit time through the sample, the end cap and the stainless steel effluent line. Transit time through the effluent line was calculated to be approximately two minutes, based on the volume of the line and an average flow rate of 6.3 ml/min for all of the experiments. Transit time through the end cap is unknown and for purposes of subsequent calculations was assumed to be zero.

The very rapid initial breakthrough in both experiments indicates that the flow rates in the largest aperture fractures were at least as fast as the initial arrival of the bacteriophage. Flow velocities in these fractures range from 70–360 m/day, calculated on the basis of measured first arrival times minus the estimated transit time through the effluent line. The steep breakthrough curves indicate there is very little dispersion of the bacteriophage and the microspheres. The steady-state concentrations of the

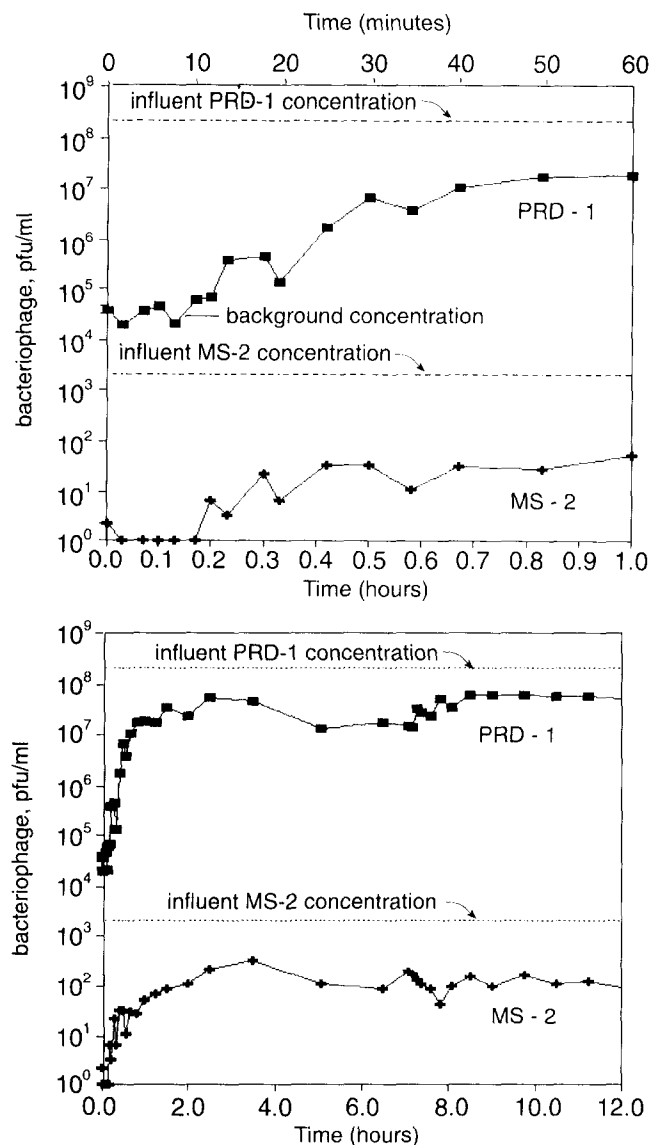


Fig. 6. Concentration of bacteriophage in effluent, experiment 2. (a) 0 to 1.0 hours; (b) 0 to 12 hours.

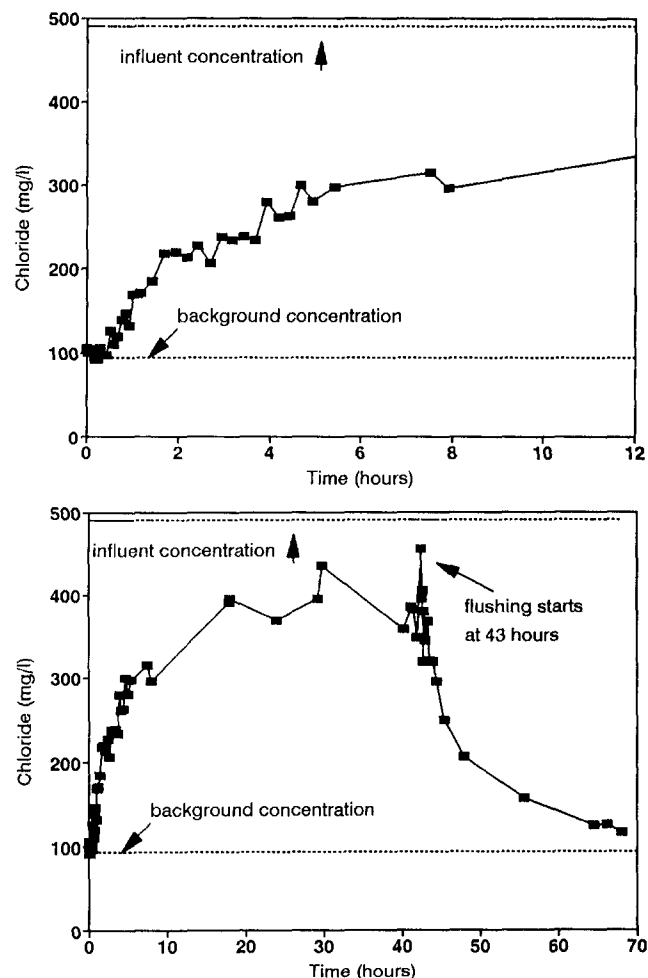


Fig. 7. Concentration of chloride in effluent, experiment 3. (a) 0 to 12 hours; (b) 0 to 70 hours.

bacteriophage after both injections are much smaller than the injection concentrations, presumably due to filtration in the fractures and to electrostatic or hydrophobic attraction to mineral or organic surfaces in the soil. High levels of entrapment of colloid-sized tracers have also been observed in previous field and laboratory experiments in fractured clay or rock (e.g., Bales et al., 1989; McKay et al., 1993b), but at present it is not possible to quantify the influence of each mechanism.

It is not known whether the bacteriophages are discharging from many fractures or only from one or two larger aperture fractures. However, it is very likely that after a steady-state concentration is reached, no additional fractures are contributing to the discharge of the bacteriophage. Based on this, the time necessary to reach steady state (60–180 minutes) for each injection should also be an indicator of the range of ground-water velocities in the contributing fractures. Using this rationale, the ground-water velocities in the slowest flowing fractures contributing to bacteriophage breakthrough would be 0.5 m in 60 to 180 minutes, or 4 to 12 m/day. In summary, the range of ground-water velocities calculated from both the first arrival and the time to reach steady state is 4–360 m/day for all of the bacteriophage injections.

Transport of Chloride

The effluent concentration versus time breakthrough curve for the chloride injection is shown on Figure 7. The first clearly detectable chloride tracer appears in the effluent approximately

30 minutes after the start of the injection. Concentrations increase rapidly for the first few hours and then increase gradually at later time. A maximum concentration of 455 mg/l, or approximately 90% of the injection concentration, is achieved at 43 hours, which is just before the clean-water flush began. The chloride concentration did not reach a steady-state value, but it appears that the effluent concentration was asymptotically approaching the injection concentration.

The initial breakthrough of the chloride is slower than observed for the colloid injections, but it is difficult to directly compare them because of the large differences in concentration ranges and detection limits. However, the chloride concentration clearly takes much longer (> 43 hours) to approach steady-state than required for the colloids (< 1 to 3 hours). This is attributed to greater diffusion of the chloride, relative to the much larger bacteriophage, into the small pores in the matrix between fractures. Previous field experiments in fractured clay (McKay et al., 1993b) using both solute (bromide) and colloidal (bacteriophage) tracers observed up to two orders of magnitude retardation of the first arrival of the bromide relative to the bacteriophage. The high flow rate and short travel distance in the lab sample would tend to reduce the separation of tracers due to differences in diffusion coefficient, but the long time required to reach steady-state is a strong indicator that matrix diffusion is retarding transport of the chloride.

Creosote (DNAPL) Injection

A total of 494 ml of the immiscible creosote mixture was injected into the sample over a one-week period (Figure 8) by increasing the reservoir pressure in small increments and measuring the subsequent creosote inflow. Pressure in the creosote reservoir was measured throughout the injection. Pressure head in the standpipe was measured only twice: first during the initial pressure increments, when creosote first began to enter the sample; and second during the final pressure increment. The pressure (in kPa) at the base of the column was calculated from these two different methods of measurement as shown on Figure 3. If there were no head losses in any of the tubing or standpipe, the two values should have been the same. The pressure values at the base of the column measured from the reservoir data, P_r ,

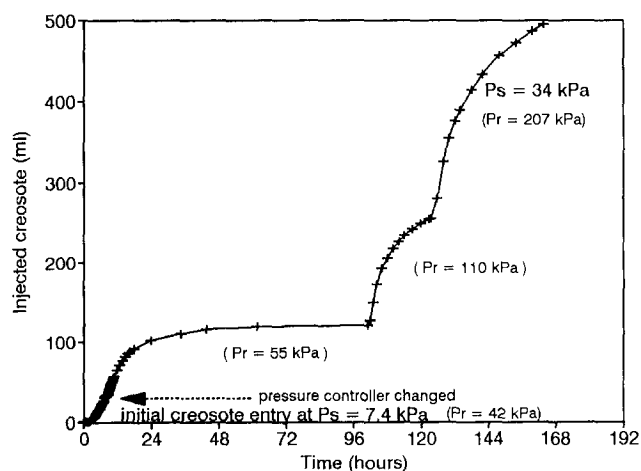


Fig. 8. Cumulative volume of creosote injected as a function of elapsed time and applied pressure (at base of column). P_r is the pressure value determined from measurements in the reservoir, and P_s is the pressure value determined from measurements in the standpipe.

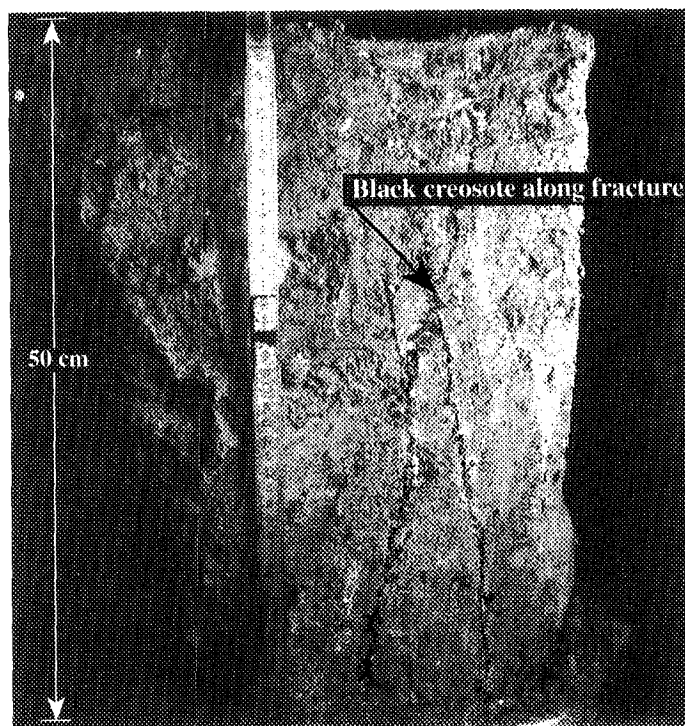


Fig. 9. Photograph of the till column after injection of immiscible creosote. The creosote stains have been darkened on the black and white photograph to enhance the contrast.

were always considerably higher than the values measured from the standpipe data, P_s . The exact causes of the differences in measured pressure are not definitively known but they may have been caused by equipment error, resistance, or partial blockage in the small (4 mm) diameter of the influent lines, the high viscosity of the creosote, and the difference in temperature between the reservoir (approximately 20°C) and the sample (8°C). The pressure values, P_s , from the standpipe are expected to be more reliable because the standpipe was connected directly to the sample end cover where the creosote entered the sample (Figure 3). Both sets of pressure measurements are shown on Figure 8.

Flow of creosote into the sample was not observed for the first few pressure increments ($P_r = 18$ to 35 kPa). As shown on Figure 8, creosote began to enter the sample at an applied pressure of $P_s = 7.4$ kPa ($P_r = 42$ kPa). After each subsequent pressure increment the rate of creosote flow increased greatly, then gradually slowed, and in some cases, stopped. The final pressure increment increased the applied pressure at the base of the sample to $P_s = 34$ kPa ($P_r = 207$ kPa). Immiscible phase creosote was not observed in the effluent line from the sample at any time during the experiment.

The till sample was then taken out of the pressure cell and carefully taken apart. Black creosote was observed across the entire influent end (the base) of the sample and was observed along many of the fractures, especially in the lower half of the sample. Creosote could be clearly seen forming an almost continuous coating on a few prominent fractures which penetrated the entire length of the sample (Figure 9). Only a small amount of creosote, a few milliliters, had exited from the top of the sample and formed a few small spots on the top surface. Creosote was not found along the rubber membrane except for a few spots of up to about 1 cm, indicating that there was no leakage around the outside of the sample.

Detailed inspection of a subsample from the interior of the column (Figure 10) showed creosote coatings on both the prominent near-vertical fractures and on several less prominent fractures as well as around many of the fine (< 1 mm) diameter root holes. As the subsample was further taken apart, many very fine root imprints were also noted along the two prominent vertical fractures. In this subsample, grey-brown reduction staining was found around almost all of the fractures or root holes which contained creosote. Furthermore, creosote was found in nearly all fractures or root holes with reduction staining. The presence of the grey-brown staining confirms that these fractures were hydraulically active at some time before the sample was taken.

Generally, the creosote-coated fractures or root holes occurred more often in the lower portion of the column. As well, creosote coatings along fractures in the lower portion of the sample tended to be more continuous than in the upper portion where they tended to follow "channels" along the fracture surface. Approximate spacing of creosote-coated fractures varied from about 5 cm in the lower portion of the column to about 20 cm in the upper portion. Since the column was placed in the pressure cell in an inverted position, this may represent an increase in fracture spacing and/or a decrease in aperture with depth. There was a similar trend towards more frequent occurrence of creosote-coated root holes in the lower portion of the column. However, the distribution of creosote may also have been influenced by the expected decrease in creosote pressure with increasing elevation in the column. For a given applied pressure at the base, this would result in the creosote being able to enter slightly smaller openings at the base than at higher elevations within the column.

Modeling and Calculation of Fracture Apertures Aperture Values from Hydraulic Conductivity

The aperture and fracture flow velocity calculations are based on the measured bulk hydraulic conductivity of the sample and the assumption that flow through the blocks of clay between fractures is negligible. The network of fractures and root holes is represented by a system of vertical orthogonal fractures as shown on Figure 11. The aperture and average velocity values calculated on the basis of the cubic law (Snow, 1969) assuming that the hydraulic conductivity of the blocks of clay between the fractures is negligible, are given by

$$2b = \left\langle \frac{K_b \cdot 2B \cdot 6 \cdot \mu}{\rho \cdot g} \right\rangle^{1/3} \quad (1)$$

and

$$v = \frac{i \cdot (2b)^2 \cdot \rho \cdot g}{12 \cdot \mu} \quad (2)$$

where v = average flow velocity in fractures; K_b = bulk vertical hydraulic conductivity of sample; $2B$ = fracture spacing; ρ = density of water at 8°C; μ = dynamic viscosity of water at 8°C; g = acceleration of gravity; and i = vertical hydraulic gradient in column.

The choice of what fracture spacing to use in calculating hydraulic-aperture values depends, in part, on the judgment of the investigator. Fractures which are not interconnected will not contribute significantly to flow and may be excluded. As well, it is very difficult to distinguish larger aperture fractures, which can largely control flow, from smaller aperture fractures. As a result, there is always some uncertainty in calculated hydraulic

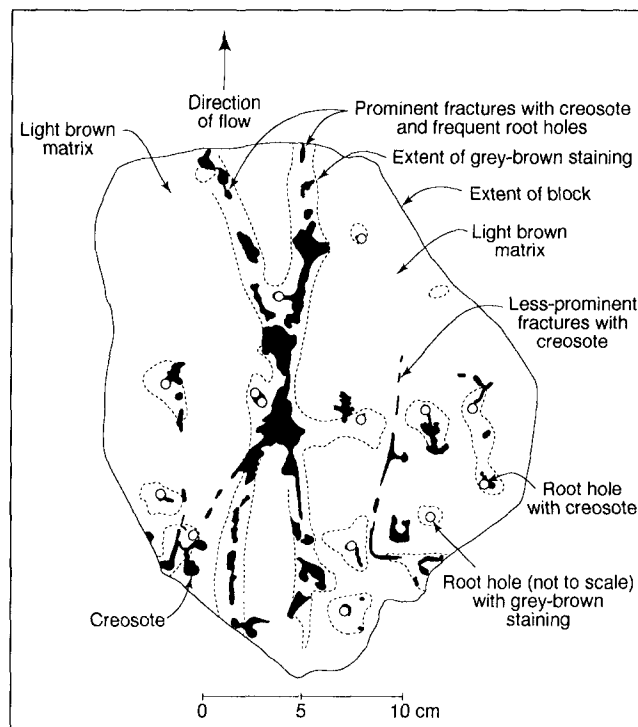


Fig. 10. Diagram of the distribution of creosote, fractures, root holes, and staining on subsample from the center of the column.

apertures or flow velocities, so expected values are presented as ranges.

For this sample the values of aperture and average flow velocity for a fracture spacing of 0.05 m, which corresponds to the smallest commonly observed spacing of visibly oxidized or reduced fractures, are 35 μm and 1.3 m/hr. By assuming a larger spacing of 0.2 m, which approximately corresponds to the spacing of two prominently stained fractures observed in the upper portion

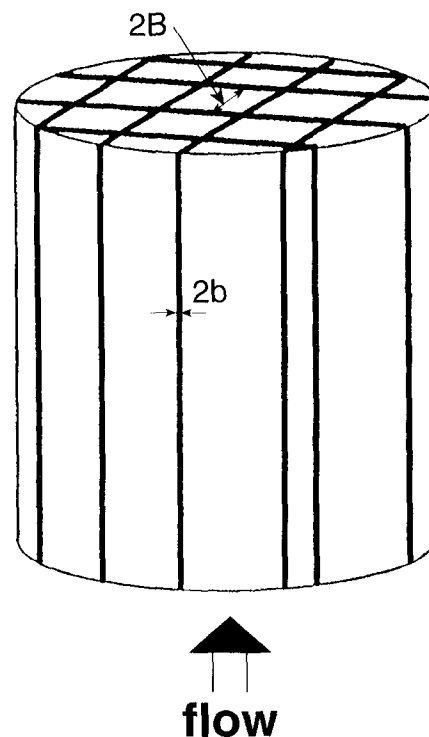


Fig. 11. Conceptual model showing fracture network used for calculation of hydraulic aperture and flow velocity.

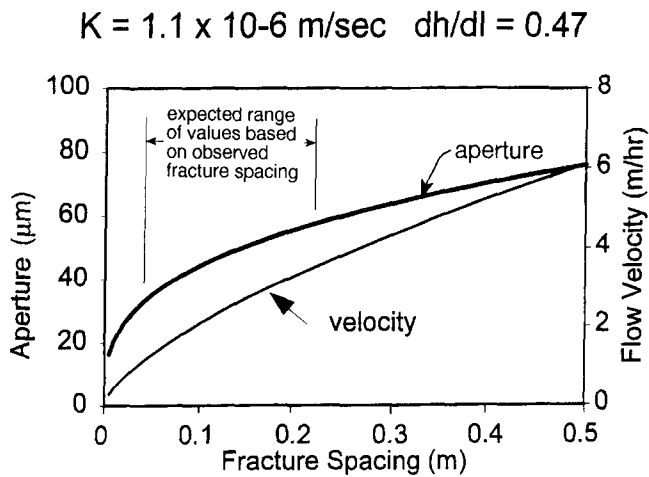


Fig. 12. Hydraulic aperture and vertical flow velocity calculated for the till column using the cubic law for a system of vertical orthogonal fractures, assuming values of average hydraulic conductivity and gradient measured during the experiments.

of the column, aperture and flow velocity values of 56 μm and 3.3 m/hr are calculated. These represent a reasonable estimate of the range of values of hydraulic-aperture and fracture flow velocity for this column, and are shown on a plot of aperture and flow velocity versus fracture spacing (Figure 12).

Aperture Values from Colloid Tracers

As previously described, the range of flow rates calculated from the observed time of first arrival and the time to reach steady-state concentration are 4 to 360 m/day (0.17 to 15 m/hr). The corresponding values of aperture calculated from equation (2) assuming a hydraulic gradient of 0.47, are 13 and 120 μm .

Aperture Values from Chloride Tracer Experiment

Solute transport simulations were performed for a 2-D, vertical cross section through the column using the CRAFLUSH model. This model is based on an analytical solution for flow and transport through a system of parallel fractures with diffusive transport in the matrix between fractures (Sudicky and Frind, 1982). The simulations were carried out for a range of fracture spacings with corresponding aperture and flow velocity values, calculated from equations (1) and (2). Values of additional required parameters, such as matrix intergranular porosity, effective diffusion coefficient, and longitudinal dispersivity were

chosen based upon values measured in this or similar deposits and are shown, with references or measurement methods, on Table 2. Most of these additional parameters, over the ranges expected for a clay-rich till, do not have nearly as great an influence on solute transport as do fracture aperture, spacing, and bulk hydraulic conductivity (McKay et al., 1993b).

The simulations were “fit” to the chloride breakthrough data by first fixing the vertical hydraulic conductivity of the sample at the lab-measured value ($1.1 \times 10^{-6} \text{ m/sec}$) and then varying fracture spacing until the simulated curves bracketed the measured data (Figure 13a). The best-fit is achieved at a fracture spacing of approximately 0.22 m, which corresponds to aperture and flow velocity values of 58 μm and 3.4 m/hr. The 0.22 m fracture spacing used in the simulations is approximately the same as the observed spacing of creosote-coated fractures in the upper portion of the sample.

A “fit” could also be achieved by fixing the fracture spacing and varying the vertical hydraulic conductivity of the sample until the simulated breakthrough matched the data (Figure 13b). The rationale for using vertical hydraulic conductivity as a fitting parameter is that the value can vary considerably within the sample. This approach is equivalent to fixing fracture spacing and then fitting the simulations by varying fracture aperture, since aperture and vertical hydraulic conductivity are directly related, as shown in equation (1). The simulations were done for a fracture spacing of 0.05 m, which corresponds approximately to the observed spacing of creosote-coated fractures in the lower portion of the sample. The best-fit was achieved with a vertical hydraulic conductivity of about $5 \times 10^{-6} \text{ m/sec}$ and the corresponding values of aperture and flow velocity are 58 μm and 3.4 m/hr.

The two approaches to fitting the simulations to the chloride breakthrough data yield essentially the same value (58 μm) of aperture as the upper end of the range of values, 35–56 μm , determined for the hydraulic-aperture. The fact that good fits can be achieved with the same aperture value (58 μm), but different fracture spacings (0.05 and 0.22 m), indicates that fracture spacing has relatively little influence on solute transport in this experiment. This is likely due to the very high flow velocity in the fractures and short length of the column. In systems with longer residence time of the water, the concentration gradients in the matrix between closely spaced fractures should be reduced due to diffusion from neighboring fractures, and hence the simulations become more sensitive to fracture spacing.

Table 2. Parameter Values Used in Calculating Hydraulic Aperture and in Modeling Chloride Transport

Parameter	Value	Units
Bulk hydraulic conductivity ¹	1.1×10^{-6}	m/sec
Hydraulic gradient ¹	0.45	dimensionless
Matrix porosity ¹	0.3	dimensionless
Effective diffusion coefficient ²	6×10^{-10}	m^2/sec
Fracture and matrix retardation factors ³	1.0	dimensionless
Longitudinal dispersivity in fractures ⁴	0	m
Fluid density ⁵	1000	kg/m^3
Dynamic viscosity of fluid ⁵	1.31×10^{-3}	$\text{kg}/\text{m}/\text{sec}$
Gravitational constant ⁵	9.81	m/sec^2

¹Measured, this study.

²Assumed value, based on summary of measured effective diffusion coefficients for chloride in a glacial till in Canada (McKay et al., 1993b).

³Assumed value.

⁴Assumed value, based on simulations of bromide migration in a fractured till in Canada (McKay et al., 1993b) and on sensitivity analyses (this study) showing that simulations were not highly sensitive to longitudinal dispersivity over the range 0 to 0.2 m.

⁵Estimated values (Daugherty and Franzini, 1977).

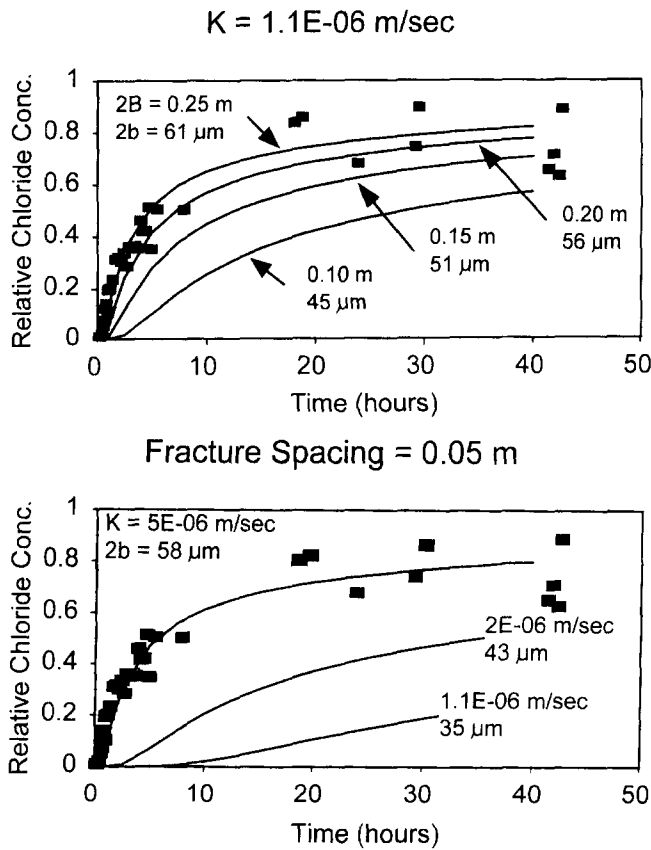


Fig. 13. Comparison of observed and simulated chloride breakthrough curves for cases with: (a) Hydraulic conductivity fixed and fracture spacing varied; (b) Fracture spacing fixed and hydraulic conductivity varied.

Aperture Values from Creosote Injection

The size of the openings of fractures and/or root holes in the till column were calculated from the entry pressure (P_E), estimated from the immiscible creosote injection, using the following two equations for parallel plate fractures and circular openings, respectively (Kueper and McWhorter, 1991):

$$\text{Fracture aperture: } 2b = 2\sigma \cos\Theta/P_E \quad (3)$$

$$\text{Root-hole diameter: } 2b = 4\sigma \cos\Theta/P_E \quad (4)$$

where $2b$ = fracture aperture or root-hole diameter; σ = the interfacial tension between the creosote and water (Table 1); Θ = the contact angle between creosote, water, and the mineral surface = 0 (assumes that water is perfectly wetting); and P_E = entry pressure = pressure difference across the creosote-water interface. The entry pressure, P_E , at any elevation within the soil column equals:

$$P_E = P_b - h_i \gamma_{\text{creosote}} - (h_{\text{out}} - h_i) \gamma_{\text{water}} \quad (5)$$

where P_b is the pressure in the creosote at the base of the column (either P_s or P_r based on measurements from the standpipe and reservoir, respectively, as shown on Figures 3 and 8); γ_{creosote} equals the specific weight of creosote (as shown in Table 1); γ_{water} equals the specific weight of water (9800 N/m³), and h_i and h_{out} are as shown on Figure 3.

When the creosote first enters the largest fracture or root hole at the base of the column, where $h_i = 0$, the P_E values calculated from P_s and P_r are 0.38 and 35 kPa, respectively, which give calculated fracture aperture values of 94 μm and

1 μm . Calculated values of root-hole diameter for the same two creosote pressure values are 188 and 2 μm .

Aperture values and root-hole diameters were also calculated for the final pressure increment of the creosote injection, which caused a small amount of creosote to exit the top of the sample (where $h_i = 0.5$ m). In this case the creosote "entry" pressure values P_E were 33 kPa and 206 kPa, calculated from P_s and P_r , respectively, resulting in calculated fracture aperture values of 1.1 and 0.2 μm . Calculated values of root-hole diameter for the same two creosote pressure values are 2.2 and 0.3 μm .

Although there is a large degree of uncertainty in the measured creosote pressures, as discussed previously, and hence in the calculated fracture apertures or root-hole diameters, the calculated values, 0.2 to 188 μm , encompass the range of fracture aperture values, 13 to 120 μm , calculated from the hydraulic data and the tracer experiments (Table 3). As previously described, creosote was observed in fine root holes (< 1 mm or 1000 μm), but it was difficult to directly measure root-hole diameters, because of their small size, and because there tended to be a "halo" of black staining around the root holes, probably due to infiltration or diffusion of the creosote.

Conclusions and Implications

In this study cubic law based "hydraulic-aperture" values provided a useful initial estimate of typical fracture aperture sizes in a column of fractured clay till. The aperture value derived from solute-transport data (58 μm) was similar, but near the upper end of the range of "hydraulic-aperture" values (35–56 μm). Values of aperture determined from the colloid-transport data had a much greater range (13 to 120 μm) than the above values and are expected to have been more strongly influenced by the largest aperture fractures, or channels within fractures.

Root holes are expected to significantly influence flow in the till at this depth (2–2.5 m) and they were not directly considered in the calculations of the "hydraulic-aperture," "solute-transport aperture," or "colloid-transport aperture" values above. It is likely that root holes are responsible for some of the observed variability in aperture measurements, and this suggests that cubic law based calculations of fracture aperture could be less appropriate for shallow soils containing numerous roots.

The bacteriophage tracers appear sufficiently mobile to be used as indicators of flow velocity in field-scale tracer experiments in fractured Danish tills. The experiments also indicate that tills, if fractured, may not provide sufficient protection against downward migration of some microbial contaminants. For example in areas where tills are thin and highly fractured

Table 3. Comparison of Fracture Apertures Calculated from Hydraulic Conductivity, Tracer Experiments, and Creosote Entry Pressure

	Fracture aperture μm	Root-hole diameter μm
<i>Calculations based on:</i>		
Measured hydraulic conductivity		
– fracture spacing = 0.05 m	35	—
– fracture spacing = 0.20 m	56	—
Chloride breakthrough	58	—
Colloid breakthrough	13 to 120	—
Creosote injection		
– based on P_s	1 to 94	2 to 188

there is potential for microbial contaminants from septic fields, manure piles, or leaking sewer pipes to rapidly migrate through the tills into an underlying aquifer.

The experiment clearly showed that the immiscible creosote mixture was capable of entering many of the visibly stained fractures and root holes in the sample under relatively low pressures corresponding to a few meters or less of pooled creosote. Because of difficulties in reliably measuring pressures in the creosote it was not possible to accurately measure values of fracture aperture or root-hole diameter, but the experiment did indicate the feasibility of applying this method to fractured clay till. The experiment also indicates that highly viscous DNAPL's such as creosote and coal tar are likely capable of penetrating fractured till deposits in the field. This is consistent with a recent survey of ground-water contamination at old coal gasification plant sites in Denmark where dissolved organic contaminants from coal tar have been found in aquifers that were overlain by till (Kiilerich and Arvin, in press). It is likely that immiscible-phase migration of coal tar and creosote compounds through the fractures in the till was one of the principal pathways for contamination of the underlying aquifers in Denmark.

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References

- Adams, M.H. 1959. Bacteriophages. Interscience Publication, New York.
- Bales, R.C., C.P. Gerba, G.H. Grondin, and S. L. Jensen. 1989. Bacteriophage transport in sandy soil and fractured tuff. *Applied and Environmental Microbiology*. v. 55, pp. 2061-2067.
- Bales, R.C., S.R. Hinkle, T.W. Kroeger, K. Stocking, and C.P. Gerba. 1991. Bacteriophage adsorption during transport through porous media: Chemical perturbations and reversibility. *Environmental Science and Technology*. v. 25, pp. 2088-2095.
- Daugherty, R.L. and J.B. Franzini. 1977. Fluid Mechanics with Engineer-

- ing Applications. 7th ed. McGraw-Hill, Inc., New York, NY. 564 pp.
- Fredericia, J. 1990. Saturated hydraulic conductivity of clayey tills and the role of fractures. *Nordic Hydrology*. v. 21, pp. 119-132.
- Grisak, G. E. and J. F. Pickens. 1980a. Solute transport through fractured media, 1. The effect of matrix diffusion. *Water Resources Research*. v. 16, pp. 719-730.
- Grisak, G. E., J. F. Pickens, and J. A. Cherry. 1980b. Solute transport through fractured media, 2. Column study of fractured till. *Water Resources Research*. v. 16, pp. 731-739.
- Houmark-Nielsen, M. 1994. Late Pleistocene stratigraphy, glaciation chronology and middle Weichselian environmental history from Klintholm, Mn, Denmark. *Bulletin of the Geological Society of Denmark*, v. 21, part 2, pp. 181-202.
- Jørgensen, P.R. and N. Foged. 1994. Pesticide leaching in intact blocks of clayey till. Thirteenth Intl. Conference on Soil Mechanics and Foundation Engineering, New Delhi, India, January 5-10, pp. 1661-1664.
- Jørgensen, P.R. and J. Fredericia. 1992. Migration of nutrients, pesticides and heavy metals in fractured clay till. *Geotechnique*. v. 42, pp. 67-77.
- Jørgensen, P.R. and N.H. Spliid. 1992. Mechanisms and rates of pesticide leaching in shallow clayey till. In: *Integrated Soil and Sediment Research: A Basis for Proper Protection*. Eds. H.J.P. Eijsackers and T. Hamers. Kluwer Academic Publishers. pp. 247-252.
- Kiilerich, O. and E. Arvin. In press. Groundwater contamination from creosote sites. Accepted for publication in *Ground Water Monitoring & Remediation*.
- Kueper, B.H. and D.B. McWhorter. 1991. The behavior of dense, nonaqueous phase liquids in fractured clay and rock. *Ground Water*. v. 29, pp. 716-728.
- McKay, L.D., J.A. Cherry, and R.W. Gillham. 1993a. Field experiments in a fractured clay till: 1. Hydraulic conductivity and fracture aperture. *Water Resources Research*. v. 29, pp. 1149-1162.
- McKay, L.D., R.W. Gillham, and J.A. Cherry. 1993b. Field experiments in a fractured clay till: 2. Solute and colloid transport. *Water Resources Research*. v. 29, pp. 3879-3890.
- Olsen, R.H., J.S. Siak, and R.H. Gray. 1974. Characteristics of PRD-1, a plasmid-dependent broad host range DNA bacteriophage. *Journal of Virology*. v. 14, pp. 689-699.
- Snow, D.T. 1969. Anisotropic permeability of fractured media. *Water Resources Research*. v. 5, pp. 1273-1289.
- Sudicky, E.A. and E.O. Frind. 1982. Contaminant transport in fractured porous media: Analytical solutions for a system of parallel fractures. *Water Resources Research*. v. 18, pp. 1634-1642.
- Sudicky, E.A. and R.G. McLaren. 1992. The Laplace transform Galerkin technique for large-scale simulation of mass transport in discretely fractured porous formations. *Water Resources Research*. v. 28, pp. 499-514.
- Tsang, Y.W. 1984. The effect of tortuosity on fluid flow through a single fracture. *Water Resources Research*. v. 20, pp. 1209-1215.